Short questions

|!|BCESC0230400001|!| **STSE**

Complete the table below which shows some information about four kinds of infectious diseases. (6 marks)

Disease	Name of pathogen	Way of transmission	One way of prevention
AIDS	(a)	(b)	Use condoms during sexual intercourse
Influenza	Influenza virus	Through air or droplets	(c)
Malaria	*Plasmodium sp.*	(d)	(e)
Tuberculosis	*Mycobacterium tuberculosis*	(f)	long course of antibiotics

##

(a) Human immunodeficiency virus (1)

(b) Through body fluids (1)

(c) Wear a face mask when in close contact with infected people / Frequently clean hands with disinfectant especially after touching public surfaces / Or other reasonable answer (1)

(d) Through mosquito bites (vectors) (1)

(e) Avoid mosquito bites by wearing light-coloured long-sleeved clothing (or applying insect repellents or using mosquito nets and screens) / Kill mosquitoes by spraying pesticides or larvicidal oil around mosquito's habitats / Remove the potential breeding grounds of mosquitoes by clearing standing water / Or other reasonable answer (1)

(f) Through air or droplets (1)

(6 marks)

##

	Skills assessed	Marks
(a)	Understanding basic concepts	1
(b)	Understanding basic concepts	1
(c)	Understanding basic concepts	1
(d)	Understanding basic concepts	1

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(e)	Understanding basic concepts	1
(f)	Understanding basic concepts	1

|!|BCESC0230400002|!|

For each of the infectious diseases listed in ***Column 1***, select from ***Column 2*** its main way of transmission. Put the appropriate letter in the space provided. (4 marks)

Column 1			***Column 2***
Japanese encephalitis	(a)	_____	A. by air and droplets
Amoebic dysentery	(b)	_____	B. through body fluids
Pneumonia	(c)	_____	C. drinking contaminated water
Syphilis	(d)	_____	D. through mosquito bites
			E. sharing personal items with infected persons

##

(a) D (1)
(b) C (1)
(c) A (1)
(d) B (1)

_____
(4 marks)

##

	Skills assessed	Marks
(a)	Understanding basic concepts	1
(b)	Understanding basic concepts	1
(c)	Understanding basic concepts	1
(d)	Understanding basic concepts	1

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|!|BCESC0230400003|!| **STSE**

For each of the infectious diseases listed in ***Column 1***, select from ***Column 2*** the method(s) that help(s) people prevent the disease. Put the appropriate letter(s) in the space provided. (5 marks)

Column 1

Hand, foot and mouth disease (a) _____

Dengue fever (b) _____

Hepatitis A (c) _____

Column 2

A. Wash hands with liquid soap after going to the toilet and before handling food.
B. Cook food and boil drinking water thoroughly.
C. Avoid sharing towels and other personal items with others.
D. Apply insect repellents.

##

(a) A, C (1, 1)

(b) D (1)

(c) A, B (1, 1)

(5 marks)

##

	Skills assessed	Marks
(a)	Understanding basic concepts	2
(b)	Understanding basic concepts	1
(c)	Understanding basic concepts	2

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|!|BCESA0230400004|!| ★★ **SI** **NOS**

In the 19th century, many women died from infections after giving birth in hospitals. A doctor observed the death rate in two maternity wards in the same hospital for a year and collected the data shown in the table.

	Ward A	**Ward B**
Number of women who gave birth	202	198
Number of women who died from infections after giving birth	67	19

The doctor also noticed that medical students were delivering babies in ward A after they had been dissecting corpses as part of their studies, while only midwives delivered babies in ward B.

(a) Calculate the death rate of women who gave birth in ward A. Show your working. (2 marks)

(b) The doctor asked the medical students to wash their hands with disinfectant before delivering babies. In doing so, the death rate in ward A had dropped to a level similar to the death rate in ward B.

(i) Suggest an explanation for the higher death rate in ward A before hand washing was introduced. (2 marks)

(ii) Explain how hand washing contributed to the lowered death rate in ward A. (1 mark)

(c) The doctor published his results but his findings were rejected by the medical community at that time. Give ***one*** aspect about the nature of science demonstrated by this. (1 mark)

##

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(a) $67 / 202 \times 100\%$ (1)

$= 33.2\%$ (1)

(b) (i) Corpses might have pathogens. (1)

The medical students might have picked up pathogens from corpses and transferred the pathogens to women in labour. (1)

(ii) By washing hands with disinfectant, any pathogens on the hands were removed or destroyed. (1)

(c) Science is socially and culturally embedded. (1)

(6 marks)

##

	Skills assessed	Marks
(a)	Mathematical skills	2
(b)(i)	Practical skills	2
(b)(ii)	Applying concepts	1
(c)	Applying concepts	1

|!|BCESC0230400005|!| **STSE**

AIDS is an infectious disease caused by human immunodeficiency virus (HIV).

(a) State how AIDS is transmitted. (1 mark)

(b) Suggest ***two*** ways that can control the spread of AIDS. (2 marks)

##

(a) It is transmitted through body fluids (e.g. semen, vaginal secretion, blood, etc.). (1)

(b) Any ***two*** of the following: (2)

- Wear gloves when handling wounds and body fluids.
- Avoid sharing syringe, needles and other objects that may be contaminated with blood.
- Screen blood before transfusion.
- Practise safe sex and wear a condom during sexual contact.

(Or other reasonable answers)

(3 marks)

##

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	Skills assessed	Marks
(a)	Understanding basic concepts	1
(b)	Understanding basic concepts	2

|!|BCESB0230400006|!| ★ **STSE**

Read the passage below and answer the questions that follow.

Gastroenteritis

Gastroenteritis is an inflammation of the stomach and the intestine. Viral gastroenteritis is caused by viruses such as Norwalk virus and rotavirus. The main symptoms are diarrhoea and vomiting. Viral gastroenteritis is an infectious disease. It is easily spread in elderly homes and schools.

(a) How is viral gastroenteritis transmitted? (1 mark)

_____

_____

(b) Cholera is transmitted in a way similar to viral gastroenteritis.
Describe ***one*** difference between cholera and viral gastroenteritis. (2 marks)

_____

_____

(c) Suggest ***two*** measures to prevent the spread of viral gastroenteritis. (2 marks)

_____

_____

##

(a) The disease is transmitted by eating food contaminated by the virus. (1)

(b) Viral gastroenteritis is caused by viruses (e.g. Norwalk virus). (1)
Cholera is caused by the bacterium *Vibrio cholerae*. (1)

(c) Any ***two*** of the following: (2)

- Wash hands thoroughly after going to the toilet.
- Consume food that is thoroughly cooked.
- Handle vomits or faeces of patients properly.

(Or other reasonable answers)

(5 marks)

##

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	Skills assessed	Marks
(a)	Applying concepts	1
(b)	Applying concepts	2
(c)	Applying concepts	2

|!|BCESB0230500007|!| ★ **STSE**

The electron micrograph below shows a type of bacteria called *Helicobacter pylori*. Studies show that *Helicobacter pylori* is able to live inside the stomach wall of humans. This bacterium can cause inflammation of the stomach wall and is associated with stomach ulcers.



(Image source: https://commons.wikimedia.org/wiki/File:EMpylori.jpg
Credits: Yutaka Tsutsumi, M.D. Professor Department of Pathology Fujita Health University School of Medicine)

(a) Based on the above information, is stomach ulcer caused by *Helicobacter pylori* an infectious disease or non-infectious disease? Explain your answer. (2 marks)

_____

_____

(b) Suggest ***one*** way to prevent stomach ulcer caused by *Helicobacter pylori* and explain how it works. (2 marks)

_____

_____

✂

(c) Stomach ulcer caused by *Helicobacter pylori* can be treated with antibiotics. Describe ***two*** mechanisms by which antibiotics kill bacteria or inhibit their growth. (2 marks)

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##

(a) It is an infectious disease (1)
caused by the pathogen *Helicobacter pylori* which can be transmitted from person to person. (1)

(b) Use serving chopsticks and spoons at meal times (1)
because *Helicobacter pylori* can be transmitted by the infected person's saliva. (1)
OR
Cook food and boil drinking water thoroughly (1)
because *Helicobacter pylori* can be transmitted by contaminated food and water. (1)
(Or other reasonable answers)

(c) Any ***two*** of the following: (2)

- Some antibiotics prevent bacterial cell walls from forming properly during cell division.
- Some antibiotics destroy the cell membrane.
- Some antibiotics inhibit protein synthesis by binding to bacterial ribosomes.
- Some antibiotics inhibit the synthesis of nucleic acids, so the bacteria cannot reproduce.

(6 marks)

##

	Skills assessed	Marks
(a)	Applying concepts	2
(b)	Applying concepts	2
(c)	Understanding basic concepts	2

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|!|BCESC0230500008|!| **STSE**

✂

John suffered from influenza and went to see a doctor. He asked the doctor to prescribe antibiotics for him but the doctor rejected his request.

(a) Which type of pathogens cause influenza? (1 mark)

_____

(b) John took measures to prevent spreading influenza to other people. Suggest ***two*** measures that he could take. (2 marks)

_____

_____

(c) Explain why the doctor refused to prescribe antibiotics to John. (2 marks)

_____

_____

##

(a) Viruses (1)

(b) Any ***two*** of the following: (2)

- Cover his mouth and nose with tissue paper when sneezing or coughing.
- Wear a surgical mask.
- Avoid going to crowded and poorly ventilated public places.
- Clean his hands with liquid soap or disinfectants after sneezing or coughing.

(Or other reasonable answers)

(c) Antibiotics are ineffective against viruses. (1)

Indiscriminate use of antibiotics may promote the emergence of antibiotic-resistant bacteria. (1)

(5 marks)

##

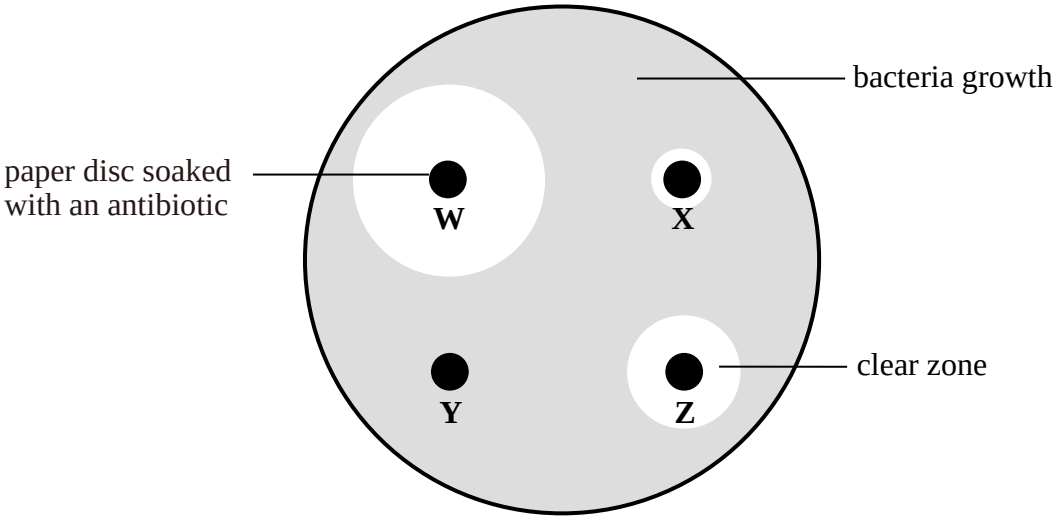
	Skills assessed	Marks
(a)	Understanding basic concepts	1
(b)	Understanding basic concepts	2
(c)	Applying concepts	2

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|!|BCESB0230500009|!| ★ **SI**

✂

Jack investigated the effectiveness of four different antibiotics (W, X, Y and Z) against a bacterium using the disc diffusion method. The diagram below shows the results.



(a) Jack used a pair of forceps to place the paper discs onto the agar plate. Each time before he picked up a paper disc, he flamed the forceps with a Bunsen flame. Explain the importance of this. (2 marks)

(b) How does the diameter of the clear zone formed around an antibiotic reflect the effectiveness of the antibiotic? (3 marks)

(c) Arrange the four antibiotics in ascending order of effectiveness in inhibiting bacterial growth. (1 mark)

(d) Suggest a control for this investigation. (1 mark)

##

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(a) To kill any microorganisms present on the forceps (1)
to avoid contaminating the agar plate. (1)

(b) Antibiotics diffuse outwards from the paper discs. (1)
The farther away from the paper disc, the lower the concentration of antibiotics. (1)
A clear zone with a greater diameter means that the antibiotic can effectively inhibit the growth of bacteria even at a low concentration. (1)

(c) Y, X, Z, W (1)

(d) Place a paper disc soaked with distilled water onto the agar plate. (1)

(7 marks)

##

	Skills assessed	Marks
(a)	Practical skills	2
(b)	Applying concepts	3
(c)	Analysing data	1
(d)	Practical skills	1

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